

CreditLens AI

Explainable credit-risk intelligence

EDA · Default prediction · Explainability · Guarded NL-to-SQL · Docker

0–1000

RISK SCORE

LOW

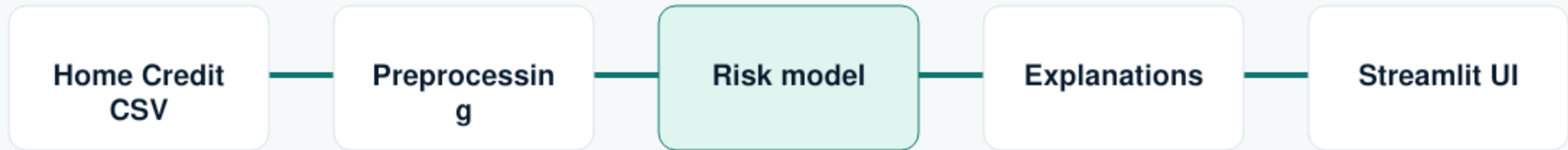
MEDIUM

HIGH

One platform connects evidence, scoring and explanation

- 1 Understand** Quality, distributions and business patterns
- 2 Predict** Default probability, score and risk band
- 3 Explain** Local drivers and readable surrogate rules
- 4 Explore** Natural language to validated SQL

The architecture keeps every decision traceable



Business question — Gemini / 5 fallbacks — SQL guard — DuckDB

One SELECT · table whitelist · mutation blocking · row limit

EDA turns data quality into five lending questions

1 Do external scores separate defaults?

2 Which income groups carry higher risk?

3 Where does credit exceed affordable income?

4 How do age and employment stability matter?

5 Can missingness reveal process weakness?

A transparent baseline handles imbalance deliberately

Split

80/20 stratified

Prepare

Impute · scale · encode

Learn

Class-weighted logistic regression

Measure

ROC-AUC · PR-AUC · confusion matrix

WHY THIS BASELINE

Fast

Auditable

Deployable

Exact metrics are generated after training—never hard-coded.

Every applicant receives probability, score and band

DEFAULT PROBABILITY

**Model estimate from
0% to 100%**

RISK SCORE

**Probability
translated to 0–1000**

RISK BAND

**Low <10% · Medium
<25% · High ≥25%**

Demonstration thresholds must be calibrated to lending loss, approval capacity and policy.

Explanations support review without claiming causality

LOCAL SHAP CONTRIBUTIONS

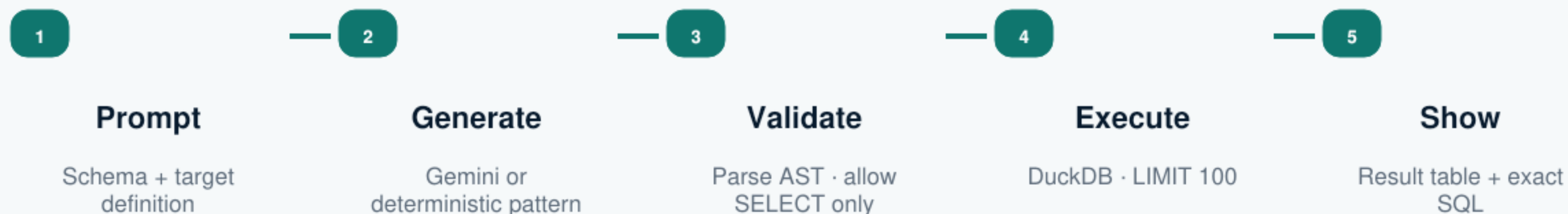


READABLE RULES

A shallow surrogate tree summarizes model behaviour for business review.

Not policy. Not causal. Validate before use.

The chatbot treats generated SQL as untrusted input



Five working questions remain available even without an API key.

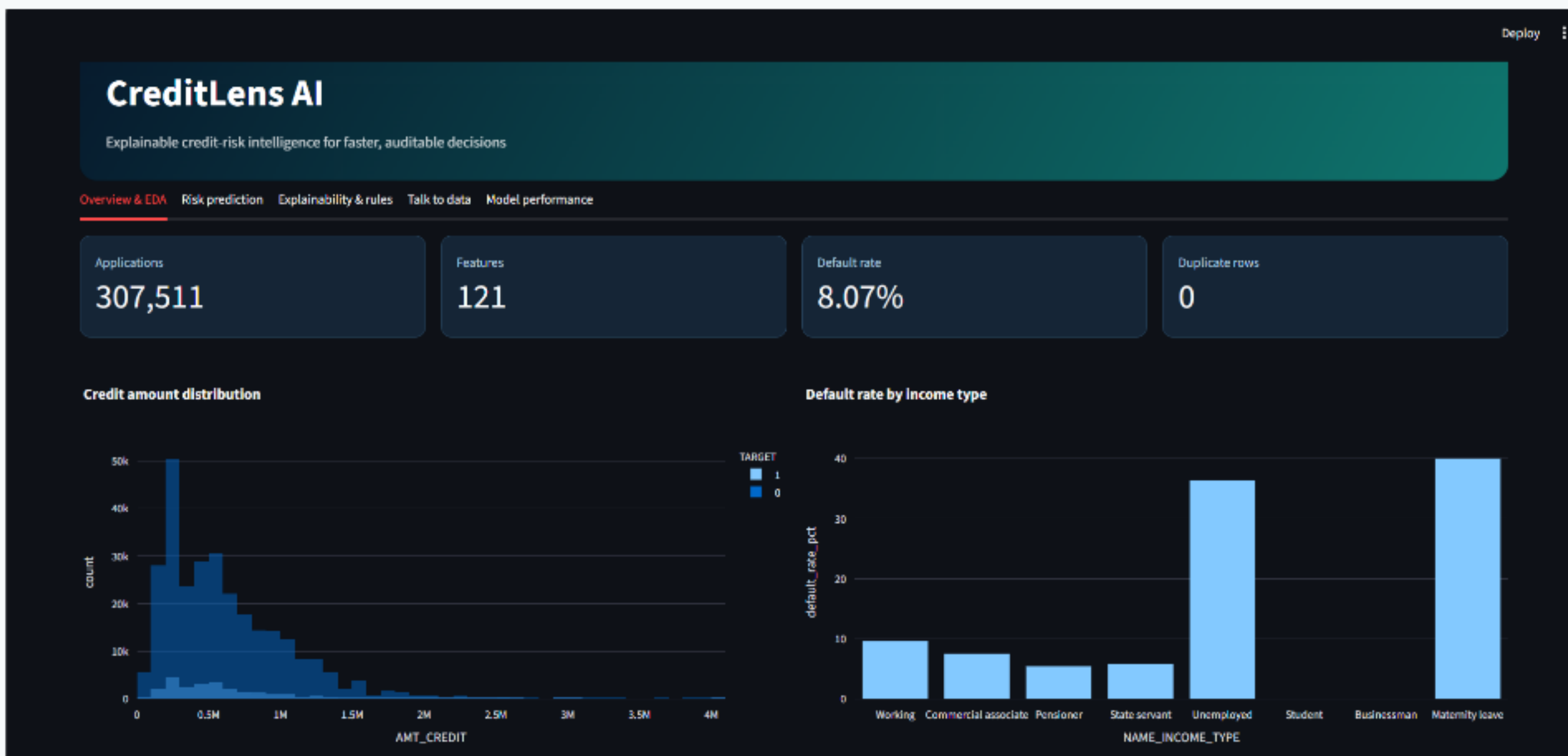
The evaluator can run the complete app with one command

docker compose up --build

Dataset mounted read-only · model artifact persisted · health check enabled

- ✓ Modular Python
- ✓ Environment contract
- ✓ Saved pipeline
- ✓ Core tests
- ✓ No dataset in Git

EDA confirms scale, imbalance and portfolio patterns



307,511 applications · 121 source features · 8.07% observed default rate

Applicant scoring produces an actionable risk band

Deploy

EXT_SOURCE_1	EXT_SOURCE_2	EXT_SOURCE_3
0.31	0.62	0.00
AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY
270000.00	1293502.50	35698.50
AMT_GOODS_PRICE	DAYS_BIRTH	DAYS_EMPLOYED
1129500.00	-16765.00	-1188.00
DAYS_ID_PUBLISH	CNT_CHILDREN	CNT_FAM_MEMBERS
-291.00	0.00	2.00
REGION_RATING_CLIENT_W_CITY	OWN_CAR_AGE	FLAG_OWN_CAR
1.00	0.00	N
FLAG_OWN_REALTY	NAME_INCOME_TYPE	NAME_EDUCATION_TYPE
N	State servant	Higher education
NAME_FAMILY_STATUS	NAME_HOUSING_TYPE	OCCUPATION_TYPE
Married	House / apartment	Core staff
ORGANIZATION_TYPE	CODE_GENDER	DAYS_LAST_PHONE_CHANGE
School	F	-828.00

Calculate risk

Default probability
61.7%

Risk score
617

Risk band
High

Example output: 61.7% default probability · score 617 · High risk

SHAP identifies the strongest drivers for one decision



Red increases predicted risk; teal reduces predicted risk. Contributions are not causal.

Surrogate rules translate model behaviour for review

Business-readable surrogate rules

```
|--- num__EXT_SOURCE_2 <= -0.21
|   |--- num__EXT_SOURCE_3 <= 0.25
|   |   |--- num__EXT_SOURCE_3 <= -8.56
|   |   |   |--- class: 1
|   |   |--- num__EXT_SOURCE_3 > -8.56
|   |   |   |--- class: 1
|   |--- num__EXT_SOURCE_3 > 0.25
|   |   |--- num__EXT_SOURCE_2 <= -1.34
|   |   |   |--- class: 1
|   |   |--- num__EXT_SOURCE_2 > -1.34
|   |   |   |--- class: 0
|--- num__EXT_SOURCE_2 > -0.21
|   |--- num__EXT_SOURCE_3 <= -8.86
|   |   |--- num__EXT_SOURCE_1 <= 0.46
|   |   |   |--- class: 1
|   |   |--- num__EXT_SOURCE_1 > 0.46
|   |   |   |--- class: 0
|   |--- num__EXT_SOURCE_3 > -8.86
|   |   |--- num__EXT_SOURCE_1 <= -1.21
|   |   |   |--- class: 0
|   |   |--- num__EXT_SOURCE_1 > -1.21
|   |   |   |--- class: 0
```

Rules summarize model behaviour and require policy-owner validation before operational use.

Depth-3 rules support audit discussion and require policy-owner validation.

Natural-language questions return validated SQL evidence

Examples: overall default rate - default rate by gender - average income by risk - top income types - credit amount by education

What is the default rate by gender?

Show the default rate by gender

	CODE_GENDER	applications	default_rate_pct
0	F	202448	7
1	M	105059	10.14
2	XNA	4	0

Validated SQL

```
SELECT CODE_GENDER, COUNT(*) AS applications, ROUND(100.0 * AVG(TARGET), 2) AS default_rate_pct FROM applications GROUP BY CODE_GENDER ORDER BY applications DESC LIMIT 100
```

What is the overall default rate?

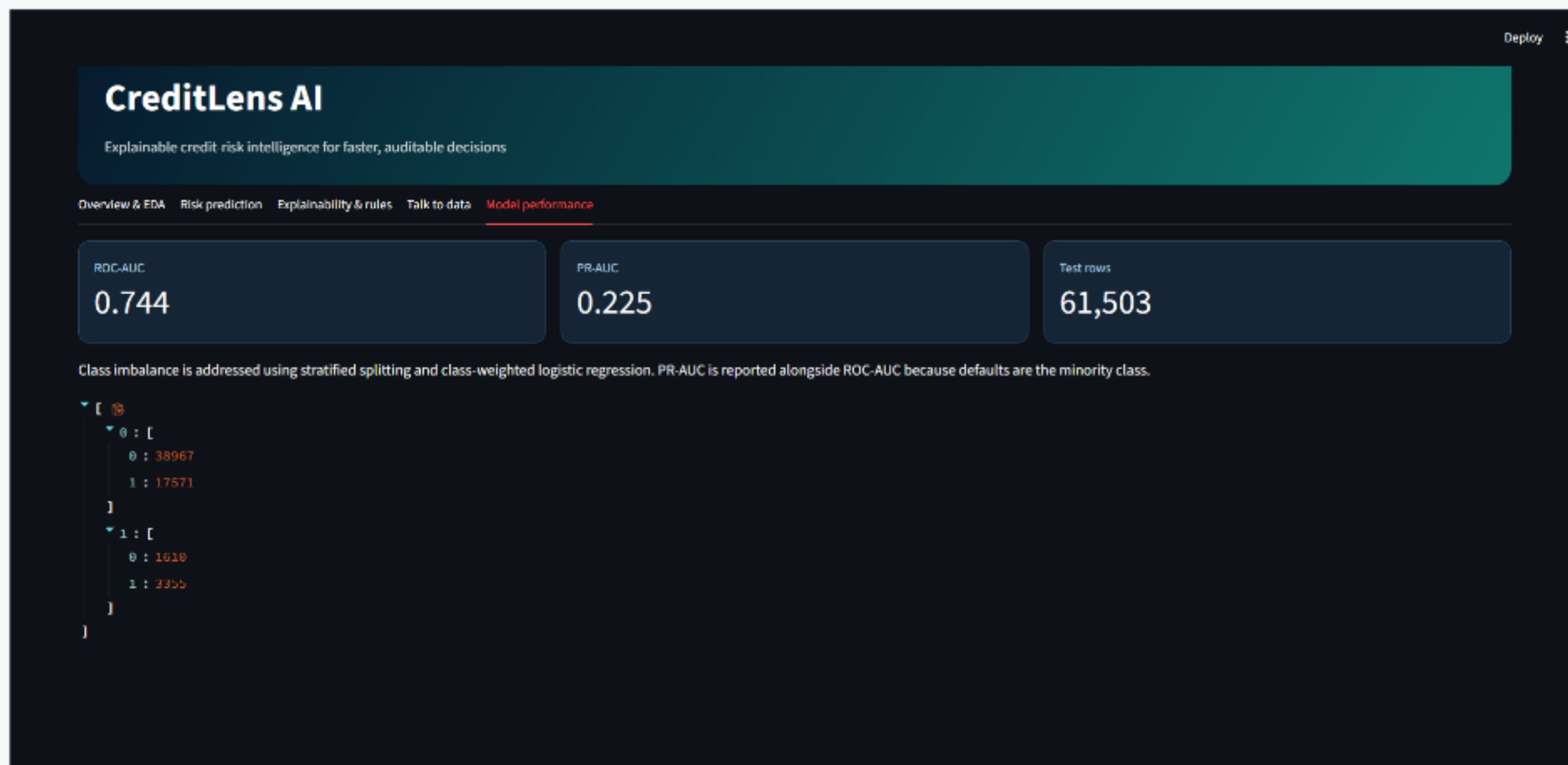
	applications	default_rate_pct
0	307511	8.07

Validated SQL

```
SELECT COUNT(*) AS applications, ROUND(100.0 * AVG(TARGET), 2) AS default_rate_pct FROM applications LIMIT 100
```

Verified patterns include overall default rate and default rate by gender.

The baseline separates risk with ROC-AUC 0.744



PR-AUC 0.225 · 61,503 test rows · stratified split and class-weighted training

The prototype is complete—and honest about production gaps

DELIVERED

- ✓ EDA + data quality
- ✓ Class-balanced risk model
- ✓ SHAP explanations + rules
- ✓ Guarded talk-to-data
- ✓ Streamlit + Docker

PRODUCTION NEXT

- Join bureau and repayment tables
- Calibrate bands to loss
- Fairness and subgroup testing
- Drift + model monitoring
- Authentication and PII controls

Decision support, with human review and policy governance.